

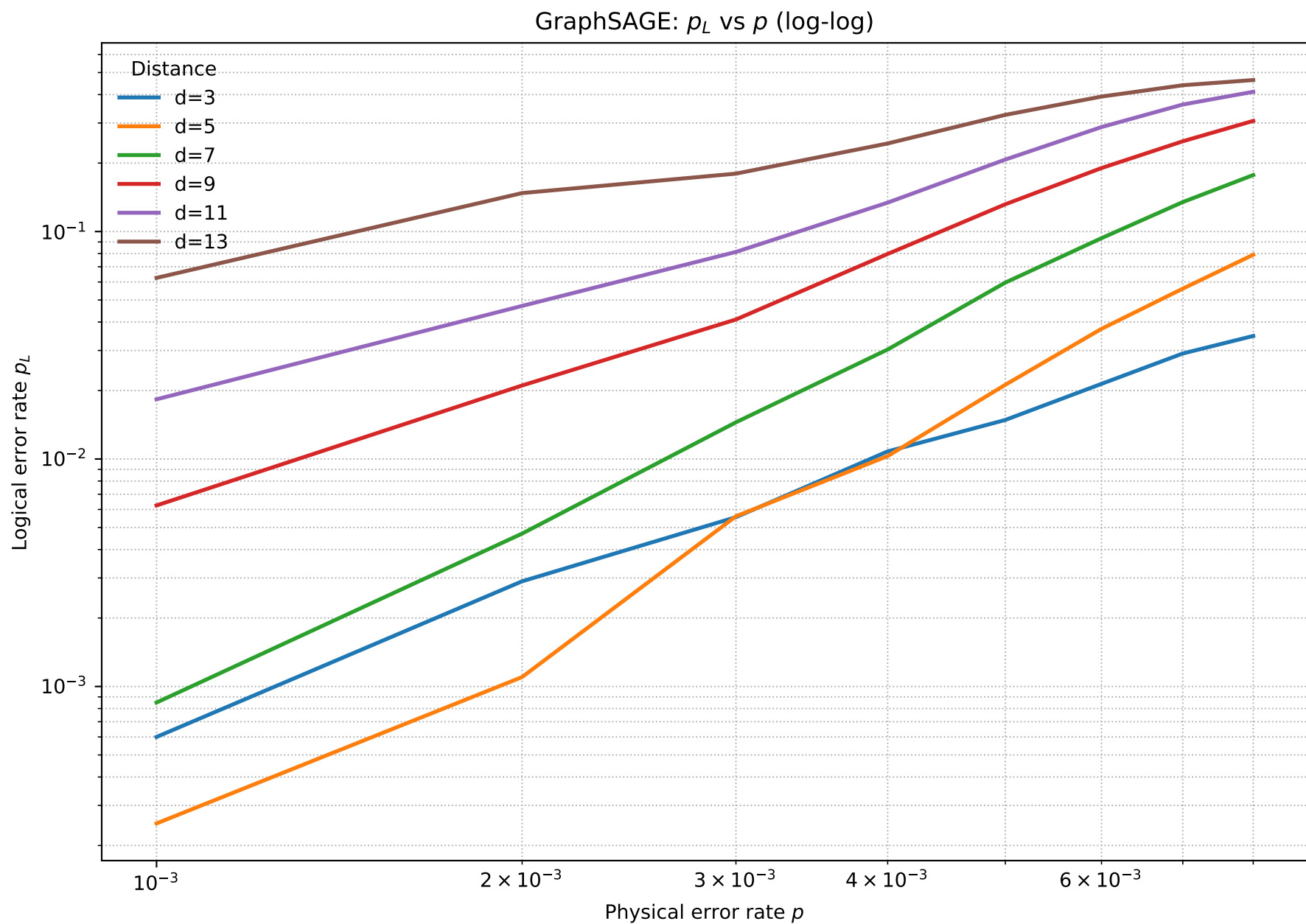


Figure A8. GraphSAGE specialist decoder logical error rate across physical error rates for distances  $d=3-13$ . Points with  $p_L = 0$  are omitted due to the log scale; lower curves indicate better suppression.

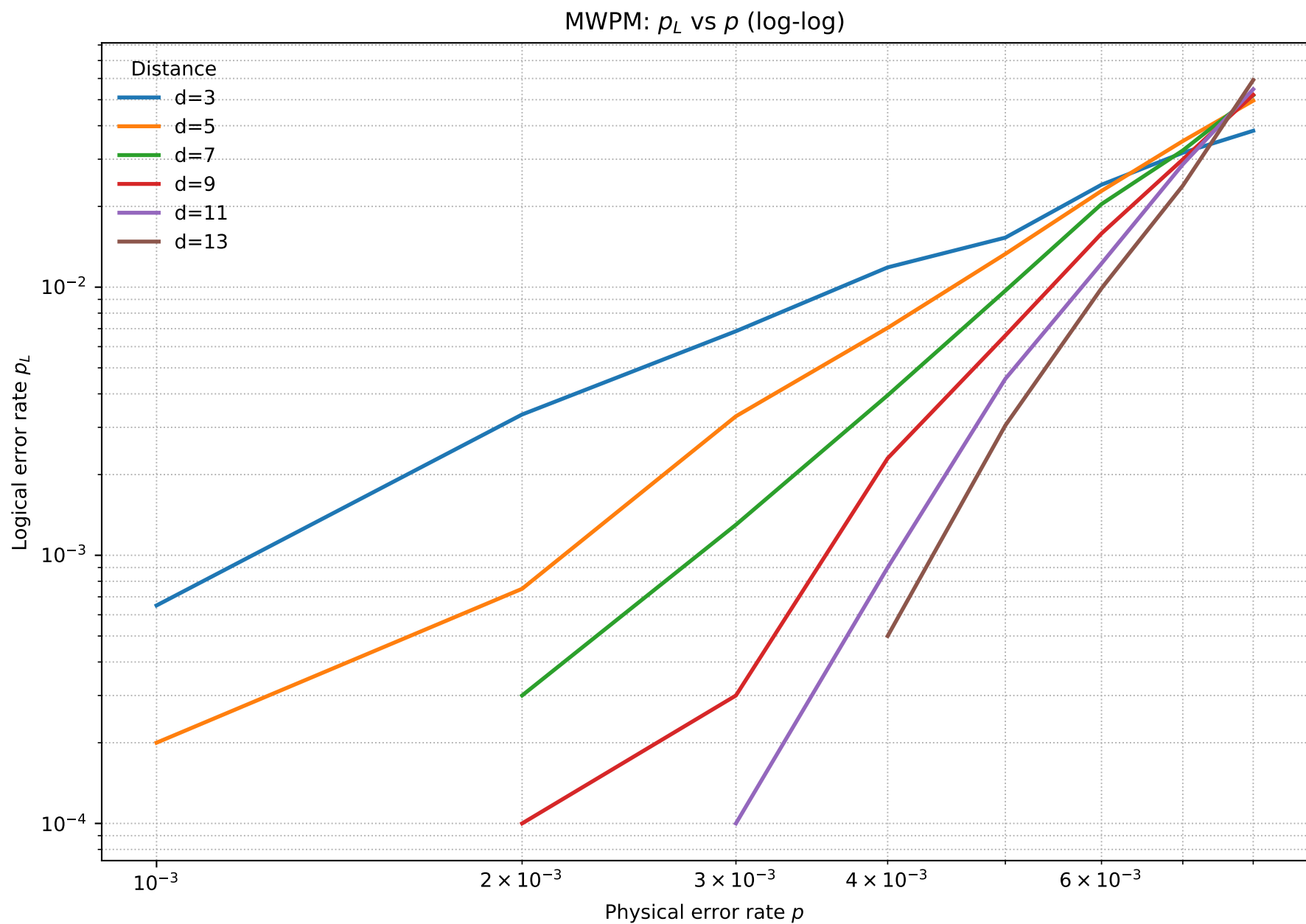


Figure A9. MWPM baseline logical error rate across physical error rates for distances  $d=3$ -13. Zero-error points are omitted on the log scale; crossover near  $p \approx 7.5 \times 10^{-3}$  marks the threshold region.

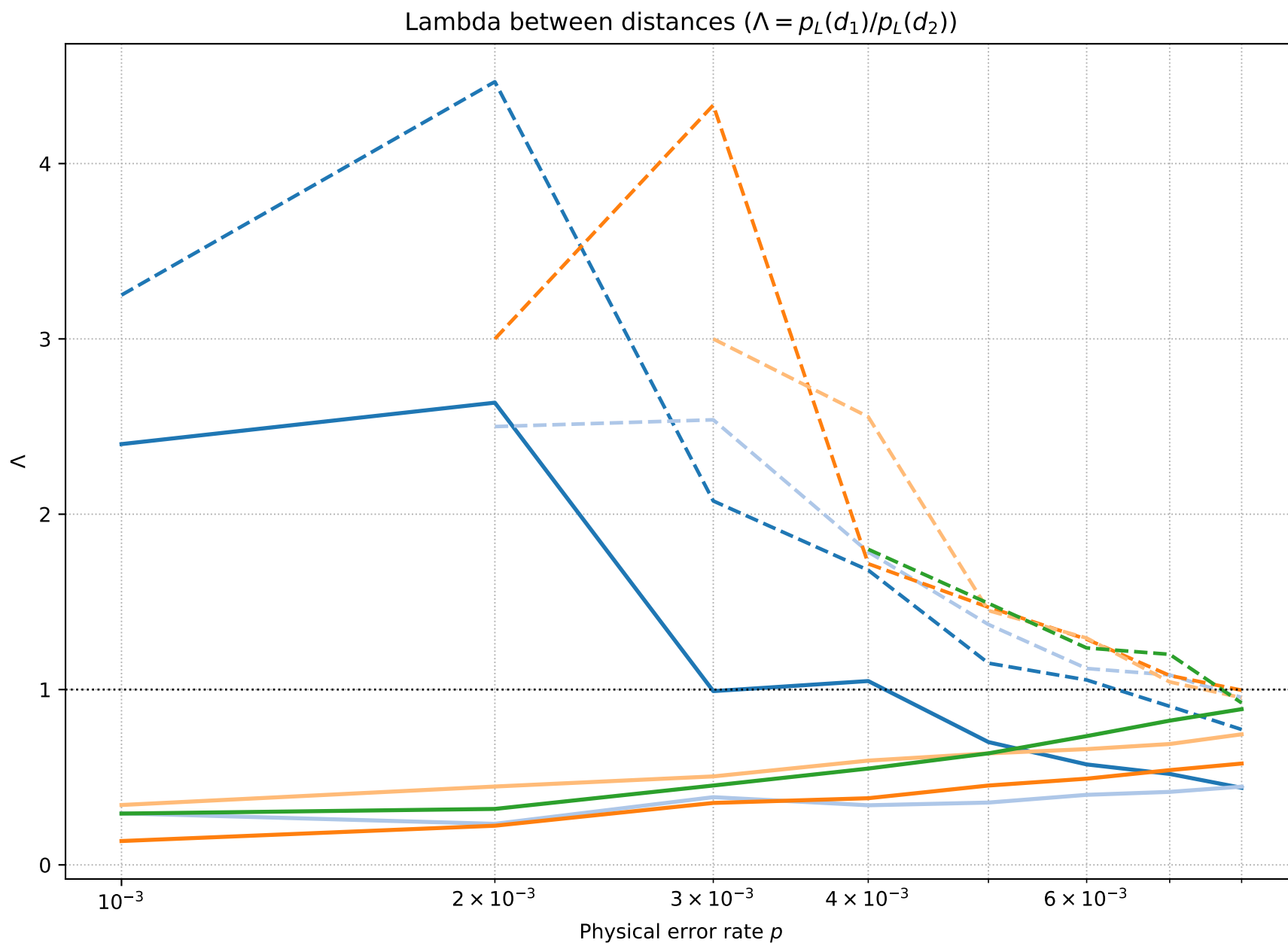


Figure A10. Error suppression factor  $\Lambda$  between adjacent distances for GraphSAGE (solid) and MWPM (dashed). Values below 1 indicate improved suppression with distance; the dotted line marks  $\Lambda = 1$ .